

AI-ASSISTED CUSTOMER COMPLAINT ASSESSMENT

Putting a Generative AI Step Between Intake and the Work Queue

A working proof of concept for medical device and pharmaceutical complaint handling

40

synthetic complaints
assessed end to end

66%

reduction in handling
time per complaint

1 vs 15

serious under-calls
AI versus keyword rules

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All figures in this document were produced by running the code in this repository.

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This document is written to be read by someone with no technical background. Every term is explained the first time it appears. Nothing assumes prior knowledge of artificial intelligence, medical device regulation, or software.

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1. Executive summary

A medical device and pharmaceutical company receives about 6,000 customer complaints a year. Some are trivial - a duplicate invoice, a late delivery. Some describe a patient who ended up in hospital. Every single one has to be read by a trained person, understood, classified, graded for seriousness, and sent to the right team. Today that takes roughly twelve minutes per complaint and it is done entirely by hand.

This project inserts one Generative AI step into that process. When a complaint arrives, the AI reads it and produces a structured draft record: what kind of complaint it is, which product it concerns, how serious it looks, which team should own it, what information is missing, and what should happen next. A qualified human being then approves, edits, or rejects that draft. The AI never closes a complaint on its own.

The single most important finding

We built two systems, not one, and tested them on the same 40 complaints.

A keyword rules engine - the cheap, obvious alternative that any IT department would propose first - graded 15 of 40 harm-related complaints as routine. It cannot detect phrases like "woke gasping" or "returned to theatre", because nobody thought to put those words on a list.

The AI arm got 1 wrong. In a regulated complaint process, that difference is not a productivity story. It is the difference between reporting a serious event on time and missing the legal deadline.

What the numbers say

Measure	Fully manual today	Keyword rules	AI + human review
Minutes spent per complaint	12.0	5.0	4.1
Complaint type identified correctly	-	72%	98%
Product identified correctly	-	98%	100%
Seriousness graded correctly	-	42%	95%
Sent to the correct team	-	65%	85%
Serious complaints graded too low	-	15 of 40	1 of 40
Records still needing a human	100%	100%	70%

On money alone the case is marginal: about \$33,200 a year of freed handler time against \$34,000 of one-off build and validation cost, so year one runs at a loss of roughly \$11,400 and the project pays back around the middle of year two. That is an honest number and the presentation should lead with it rather than hide it.

The case is not carried by the money. It is carried by the under-call figure.

2. The business problem we set out to solve

What complaint handling actually involves

When a customer, nurse, doctor, or distributor reports a problem with a medical device or a medicine, the manufacturer is legally obliged to do something specific with that report. It must be recorded, assessed for how serious it is, and - if it meets certain thresholds - reported to a regulator within a fixed number of days. In the United States this reporting duty is called MDR, short for Medical Device Reporting. The clock starts when the company becomes aware of the event, not when it gets round to reading it.

The people who do this are called complaint handlers. For each incoming complaint a handler must:

- read free-form text written by someone who is upset, in a hurry, or not medically trained;
- work out which product is being described, often from a nickname or a partial serial number;
- decide what category of problem it is;
- grade how serious it is, which is a judgement about potential harm, not a lookup;
- decide whether it might be reportable to a regulator;
- work out what facts are missing and will have to be chased;
- send it to the right internal team.

The three costs this creates

Cost	What it looks like in practice	Rough scale
Handler time	About 12 minutes of reading and re-typing per complaint, before any investigation begins.	~1,200 hours a year
Rework from misrouting	A complaint that should have gone to Pharmacovigilance sits in a Customer Service queue for days before anyone notices.	20 minutes lost per misroute
Inconsistency	Two handlers grade the same complaint differently. Trend data built on inconsistent categories cannot support a safety decision.	Unquantified, but corrosive

Why the third cost is the worst one

The first two cost money. The third costs you the ability to see a pattern.

If the same failing component is described as a "Device Malfunction" by one handler and a "Quality issue" by another, the count in each bucket is wrong, and the signal that should have triggered an investigation never rises above the noise.

3. What was actually built

This is a working system, not a slide deck. The whole comparison runs on a laptop with nothing installed beyond Python itself, because the AI output ships with the project as a saved file.

Regenerating that output is the one step that needs an internet connection and an API key - see the appendix. There are five moving parts.

Component	What it is	In plain language
The complaint data	40 synthetic complaints in a spreadsheet file	Realistic but invented reports. No real patient data anywhere.
The product catalogue	10 products - 8 devices, 2 medicines	The list the AI is allowed to choose from. It cannot invent a product.
The prompt	A versioned instruction document	The exact wording given to the AI. Kept in its own file so it can be reviewed, version-controlled and audited like any other regulated document.
The pipeline	One Python script with three commands	run - assess the complaints. evaluate - score the results. dashboard - refresh the display.
The dashboard	A single web page	Opens in any browser with no server. Shows the queue a manager would actually work from, plus a side-by-side scorecard of the two engines.

Two engines, deliberately

The project builds the same output twice. Once with the AI, and once with a keyword rules engine written to be the honest cheap alternative. This matters because the interesting question is never "does AI work?" - it is "does AI beat the thing that costs nothing?" Without that second arm, every accuracy number in this document would be unanchored.

4. The workflow, step by step, in plain language

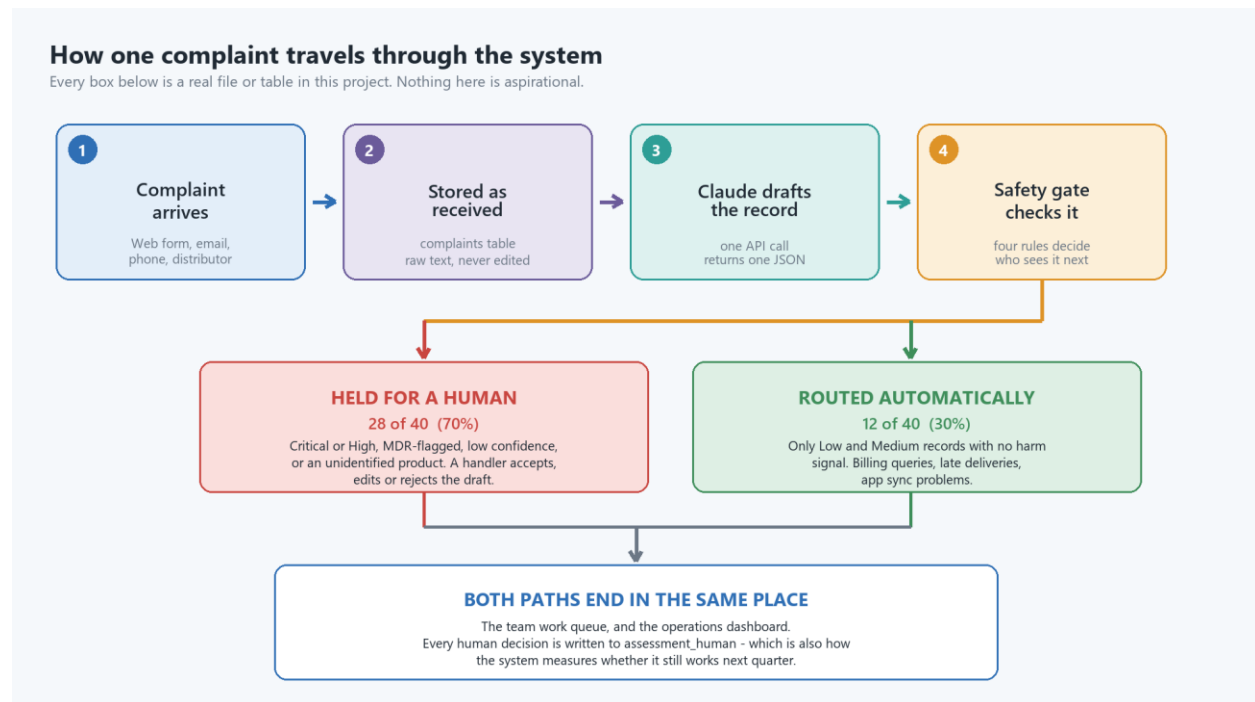


Figure 1 - The complete journey of a single complaint through the system.

Step 1 - A complaint arrives

Complaints come in through four channels: a web form on the company site, an email inbox, a telephone line where an agent types up what was said, and a portal used by distributors who sell the products in other countries. All four produce the same thing: a block of free-form text plus whatever details the reporter happened to supply.

Crucially, that text is messy. A reporter may write "the infusion pump" instead of "InfuLine 300". They may not give a lot number. They may describe two separate problems in one paragraph. The system has to cope with all of that, because that is what real intake looks like.

Step 2 - It is stored exactly as received

The raw complaint is written to a database table called complaints and is never edited. This is a regulatory requirement, not a design preference: the original words of the reporter are the legal record. Everything the system produces afterwards is stored separately, so you can always go back and see what was actually said.

Step 3 - The AI drafts a structured record

This is the Generative AI step, and it is deliberately small: one complaint in, one structured record out, in a single call. There is no conversation, no memory between complaints, and no chain of steps that could drift. Each complaint is assessed in isolation, which means the result for complaint 12 cannot be contaminated by complaint 11.

What the AI is told to produce

The AI is given three things: a set of rules about how to behave, the list of allowed answers, and the complaint itself. It must return a fixed set of fields:

Field	What it means	Why it is there
category	The type of problem, chosen from a fixed list of 10	Keeps the data joinable to the existing quality system. No invented labels.
product_id	Which catalogue product, or UNKNOWN	You cannot trend or recall a product you cannot name.
severity	Critical, High, Medium or Low	Drives the priority of everything downstream.
mdr_reportable_flag	Yes, Possible or No	Flags a potential legal reporting duty early, while the clock still has time on it.
summary	One factual sentence, 25 words maximum	Lets a handler assess from the queue without opening the full record.
missing_info	A list of facts the handler must chase	This is the real bottleneck in complaint handling - chasing facts, not typing.
suggested_next_action	One concrete next step	Turns a complaint into a task somebody can act on.
route_team	Which internal team owns it	Removes the routing guesswork that causes days of delay.
confidence	The AI's own certainty, from 0 to 1	Feeds the safety gate. Low confidence forces human review.
needs_human_review	True or false	The output of the safety gate described in section 7.

Three rules that shape the AI's behaviour

These are worth understanding because they are the difference between a demo and something that could survive an audit.

Rule 1 - Never invent anything

If a fact is not in the complaint text, the AI must write "Not stated" and add the field to the missing-information list.

A hallucinated lot number in a regulated record would itself be an audit finding. The system is designed so that the AI's ignorance is visible rather than papered over.

Rule 2 - When in doubt, grade it higher

The AI is explicitly instructed to lean towards the more serious grade when the text is ambiguous. This is a deliberate, asymmetric choice. Over-grading costs a few minutes of a handler's time. Under-grading can mean a missed regulatory report. The two errors are not equally expensive, so the system is not tuned to treat them equally.

Rule 3 - Answer only in structured data

The AI returns machine-readable data, not prose. It cannot ramble, hedge, or add commentary. If it cannot produce a valid record, the pipeline fails loudly rather than writing something half-formed into a regulated system.

Step 4 - The safety gate decides who sees it next

Every drafted record is tested against four conditions before it goes anywhere. This is covered in full in section 7, because it is the heart of the design.

Step 5 - A human accepts, edits, or rejects

Records that trip the gate land in a review queue, ordered with the most serious first. A handler opens the queue, sees the AI's draft alongside the original complaint, and makes a decision. That decision is written to a separate table called `assessment_human`.

This is not just an approval step. It is the measurement system. Because the AI's draft and the human's final answer are both stored, the organisation can calculate, every week, how often the AI was right - and notice immediately if that starts to slip after a model update.

Step 6 - The dashboard

Both paths - reviewed and auto-routed - end in the same team work queue and the same dashboard. A manager sees the volume, the severity mix, the routing load across teams, a side-by-side scorecard of the two engines, and the live review queue with the highest-priority items at the top.

5. The data we used, and why we kept it small

The project uses 40 synthetic complaints. That number was chosen deliberately, and the reasoning is worth stating.

Why synthetic

Real complaint text contains patient names, dates of birth, clinical details and contact information. Using real records in a student project would be both unlawful and unnecessary. Every complaint in this dataset was written to look and read like a real one - including the vagueness, the missing lot numbers, and the emotional tone - but none of it describes a real person.

Why only 40

Forty is large enough that the accuracy percentages mean something and small enough that a human being can read every single record and check the system's answer by hand. That second property matters more than it sounds. With 40 records, every claim in this document is verifiable by a reader in an afternoon. With 4,000 you would have to take the numbers on trust.

What is in the dataset

Ingredient	Detail
Complaints	40 records, spanning June 2026, across four intake channels
Reporter types	Patients, clinicians, clinic administrators, distributors
Regions	US Northeast, Midwest, West, South; EU DACH and Nordics
Products	8 devices and 2 prescription medicines, plus an UNKNOWN entry
Severity mix	6 Critical, 15 High, 9 Medium, 10 Low
The answer key	A separate file giving the correct category, product, severity and team for all 40, used to score both engines

The deliberate difficulty in this dataset

Several complaints describe genuine harm without using any of the words a keyword system would look for.

One reporter writes that a patient "woke gasping". Another says a patient was "returned to theatre". A third describes a neonate who became "fluid overloaded".

None of those phrases contain the words hospital, emergency, injury or death - and that is precisely the point. This is what real intake text looks like, and it is where keyword systems quietly fail.

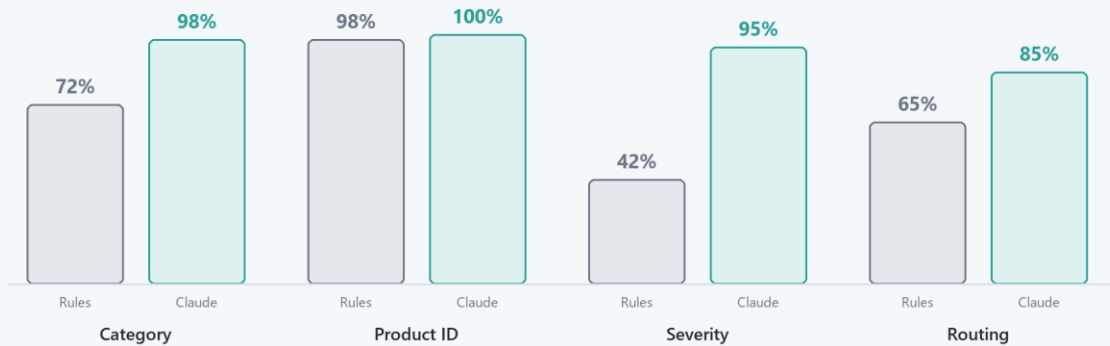
6. Where Generative AI genuinely adds value

It would be easy - and dishonest - to claim AI improves everything. It does not. The table below separates the steps where a rules engine is perfectly adequate from the steps where it structurally cannot compete.

Step in the process	Could simple rules do it?	What the AI adds
Identify the complaint type	Partly - 72% correct	Reads intent rather than keywords. "Nobody explained how to prime it" is a training gap, not a device fault.
Identify the product	Mostly - 98% correct	Resolves nicknames such as "Glucotrack meter" or "the infusion pump" against the catalogue, and correctly returns UNKNOWN when it genuinely cannot tell.
Grade the seriousness	No - 42% correct	This is the decisive one. The AI infers harm from narrative. A rules engine has no keyword for "woke gasping".
Spot what information is missing	No - cannot do it at all	Rules can only see what is present. The AI names what is absent: lot number, event date, patient outcome. This is the actual bottleneck in complaint handling.
Write a usable summary	No	A 25-word factual line, so a handler can assess from the queue without opening the record.
Suggest the next action	No	Converts a complaint into a task somebody can pick up.
Route to the right team	Partly - 65% correct	Recognised a misdelivered package that exposed another patient's details as a privacy incident rather than a shipping problem.

Where the AI beats a keyword rules engine

Same 40 complaints, same answer key, two different engines. Taller is better.



THE NUMBER THAT MATTERS MOST

Serious severity under-calls: rules engine 15 of 40 Claude 1 of 40

An under-call is a harmful complaint filed as routine.
That is how a reportable event misses its filing deadline.

Figure 2 - Accuracy of both engines on the same 40 complaints, with the safety measure called out.

A worked example, so the difference is concrete

Why keyword rules fail: a worked example (complaint C-1031)

This is a real record in the dataset. Read the patient's words, then look at what each engine did with them.

"Unit shut off twice overnight without alarm; patient woke gasping and has stopped using the device." - AeroRest CPAP, patient report

KEYWORD RULES ENGINE

Searches for "hospital", "emergency room", "died".
Finds none of them. The words "woke gasping" are
not in any list a person thought to write.
Graded LOW - routed as a routine query.

CLAUDE

Reads the sentence. A breathing-support device that
stops without alarming, on a sleeping patient, is a
failure of a life-sustaining therapy.
Graded HIGH - MDR flagged, sent to a human.

Figure 3 - The same complaint, read by both engines, with opposite results.

This single record is the clearest argument in the whole project. A CPAP machine is a breathing-support device used overnight. The complaint says it shut off twice without sounding an alarm and the patient woke gasping.

The keyword engine searched for words on a list, found none of them, and graded the complaint Low - the same grade it would give a query about an invoice. The AI read the sentence, understood that a breathing device failing silently on a sleeping patient is a failure of a

life-sustaining therapy, graded it High, flagged it as potentially reportable, and sent it to a human.

7. The human review point - the safety gate

Every serious AI deployment needs a clear answer to one question: at what precise moment does a human being become responsible? In this system that moment is defined in code, and it is not optional.

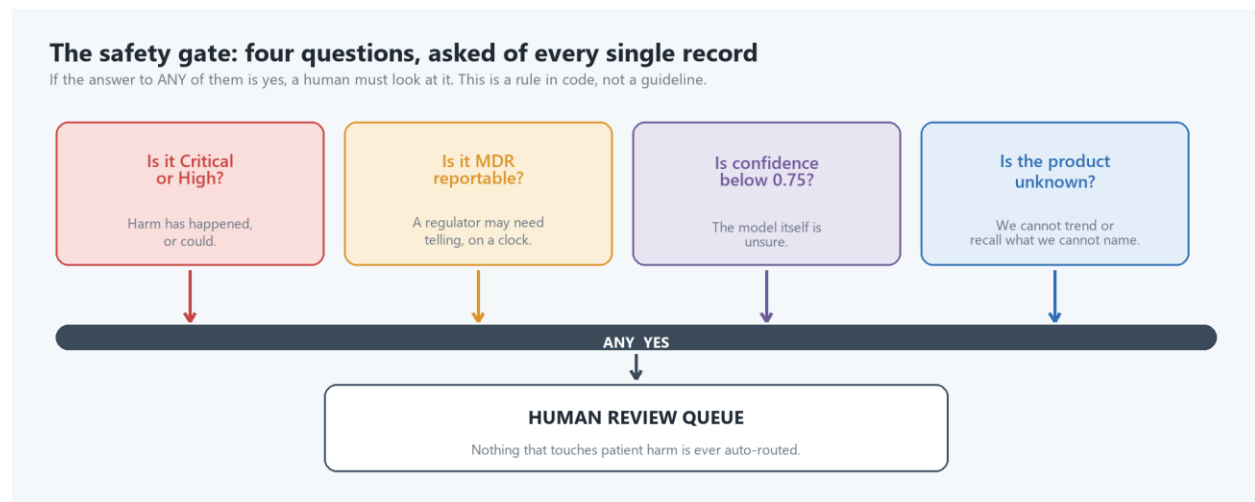


Figure 4 - The four conditions that force human review. Any one of them is enough.

Why these four conditions

Condition	Why it forces a human
Severity is Critical or High	Harm has happened or could. This is the category where being wrong is most expensive, so a machine is never the last word.
Reportable flag is Yes or Possible	A regulatory clock may already be running. A person must confirm the assessment before the deadline is assumed.
Confidence is below 0.75	The AI is telling you it is unsure. Ignoring that signal would waste the most useful thing the model produces.
Product could not be identified	An unidentified product cannot be trended, investigated or recalled. Someone has to resolve it before the record is of any use.

What this means in numbers

28 of the 40 complaints (70%) were held for human review.

The remaining 12 (30%) were routed automatically - and every one of them is a low-severity billing query, delivery delay, or app synchronisation problem with no harm signal of any kind.

Nothing that touches patient harm is ever auto-routed. That is a property of the code, not a

promise in a policy document.

The review is also the monitoring system

Every accept, edit and reject is stored. Comparing the AI's draft against the human's final answer produces an ongoing agreement rate. If that rate falls - after a model version change, or because the product range shifted - it shows up in the data within days. The rule agreed in this design is that if agreement drops below 85% and cannot be recovered, automatic routing is switched off and everything reverts to full human review.

8. Business metrics: what we measure and why each one matters

Two metrics decide whether this project was worth doing. Several others support them. Each one is defined below with its purpose, how it is calculated, what it currently reads, and what would make it meaningless.

Primary metric 1 - Handling time per complaint

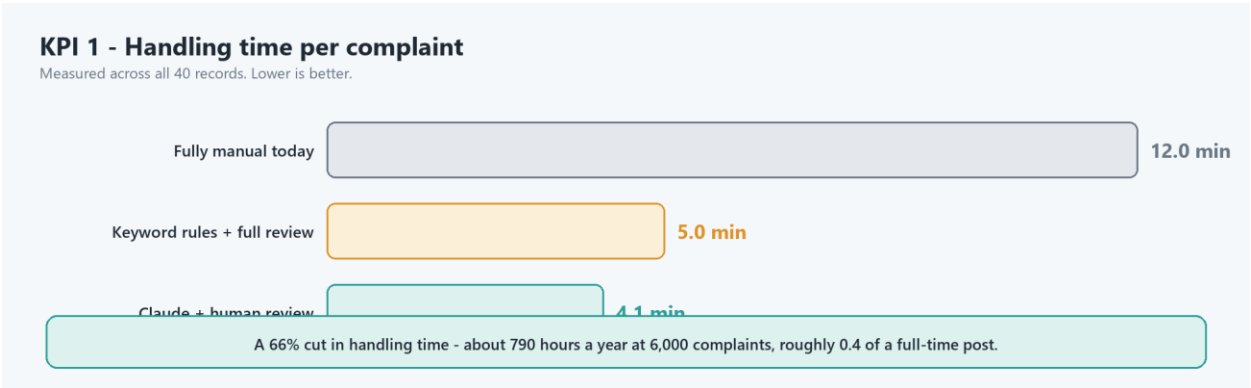


Figure 5 - Average handling time, measured across all 40 records.

Aspect	Detail
What it measures	The average number of minutes a handler spends taking one complaint from arrival to correctly routed.
How it is calculated	Reviewed records are counted at 5 minutes (read the draft, check it, accept or edit). Auto-routed records are counted at 2 minutes (spot-check only). $70\% \times 5 + 30\% \times 2 = 4.1$ minutes.
Current reading	4.1 minutes, against 12.0 fully manual - a 66% reduction.
Why it matters	It converts directly into money and into capacity. At 6,000 complaints a year it frees roughly 790 hours, about 0.4 of a full-time post, worth about \$33,200.
What would falsify it	If real manual intake turns out to be 6 minutes rather than 12, the benefit halves and the project does not clear its cost. This assumption should be measured with a stopwatch on 30 real complaints before anyone funds the build.

Primary metric 2 - Serious severity under-call rate

This is the metric with veto power over the whole project.

Aspect	Detail
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What it measures	The percentage of complaints that the AI graded Medium or Low, and that the human reviewer then had to upgrade to High or Critical.
How it is calculated	Count the records where the AI's severity is lower than the agreed correct severity, and the correct severity is High or Critical. Divide by the total.
Current reading	1 of 40 for the AI. 15 of 40 for the keyword rules engine.
Why it matters	An under-called complaint is a harmful event filed as routine. In a regulated process this is how a reportable event misses its filing deadline - and one late report can cost more in remediation and regulatory exposure than the system saves in a decade.
Target	Zero. It is not currently zero, which is exactly why the human review step exists and why auto-routing is restricted to low-severity records.
What would falsify it	If the reviewer population disagrees with the answer key used here, the metric is measuring the wrong thing. It needs re-baselining against two independent reviewers on real data.

Supporting metrics

Metric	Reading	Why it is tracked
Category accuracy	98% vs 72%	Wrong categories corrupt trend data, which is what safety investigations are built on.
Product identification	100% vs 98%	You cannot recall or investigate a product you cannot name.
Routing accuracy	85% vs 65%	Every misroute costs about 20 minutes of rework and days of delay. Note this is NOT claimed as a saving - see the honest note below.
Human review rate	70%	If this never falls, the time saving never materialises. It is the dial that converts accuracy into money.
Average AI confidence	0.88	A leading indicator. If it drifts downward the model is struggling before accuracy visibly drops.
Agreement rate over time	To be tracked live	The early-warning system for model or prompt drift. Below 85%, auto-routing is disabled.

An honest correction we made to our own business case

Our first draft claimed a \$7,600 annual saving from better routing. That claim was wrong and we removed it.

The arithmetic: manual intake is assumed to misroute 15% of the time. The AI measured 85% routing accuracy - which is also a 15% misroute rate. The difference is zero. The AI beats the rules engine at routing, but it does not beat a human, and the business case is measured against humans.

Removing that line moved year one from a \$3,800 loss to an \$11,400 loss. We would rather

present the smaller number and have it survive scrutiny.

The full financial picture

Line	Amount
Handler time freed: 790 hours x \$42 per hour	\$33,200 per year
Rework avoided	\$0 - not claimed, see above
Gross annual benefit	\$33,200
Build cost, one-off	\$22,000
Validation and documentation, one-off	\$12,000
AI model and database, annual	\$600
Prompt maintenance and monitoring, annual	\$10,000
Year 1 net	-\$11,400
Year 2 onward	+\$22,600 per year
Payback point	Around the middle of year 2

Note that the AI itself costs almost nothing: about one cent per complaint, under \$200 a year. The expense in a regulated environment is never the computing. It is the validation, the version control, and the human review capacity you must keep staffed.

9. Proof of concept: results and screenshots

Everything below is a screenshot of the working system, taken from the dashboard that ships with this project. It opens in a browser with no server and no installation.

The headline panel

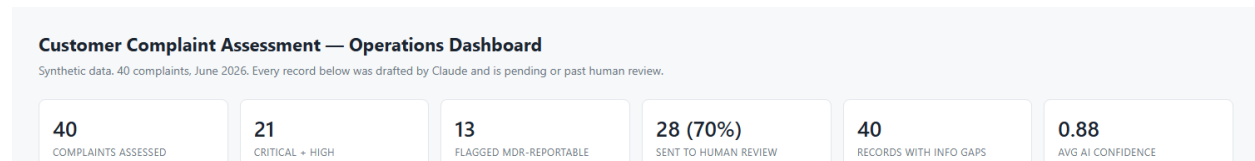


Figure 6 - Live summary across the 40 assessed complaints.

Reading left to right: 40 complaints assessed; 21 of them graded Critical or High; 13 flagged as potentially reportable to a regulator; 28 (70%) held for human review; all 40 have at least one piece of missing information the handler must chase; and the AI's average confidence in its own answers is 0.88.

The analytical views

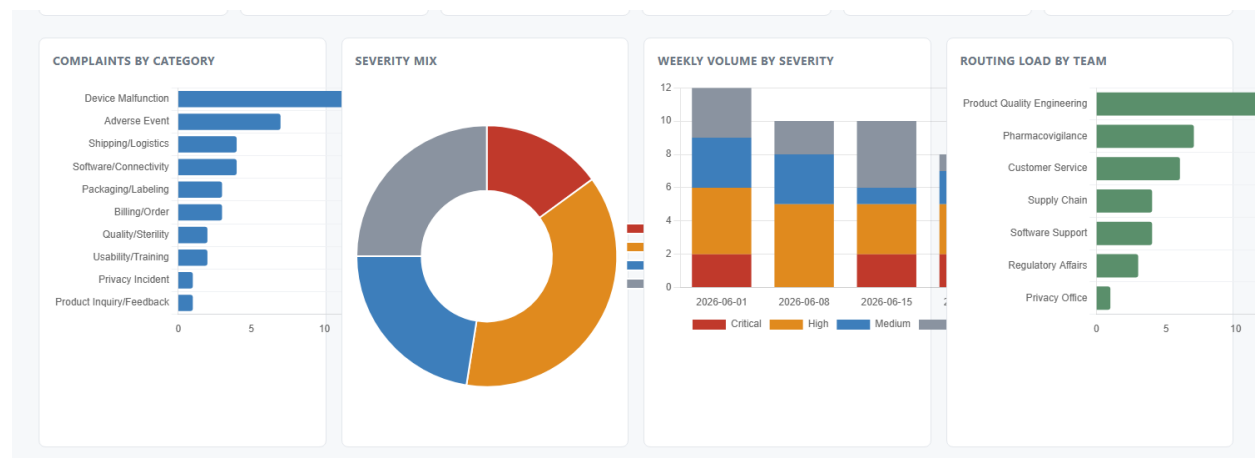


Figure 7 - Category mix, severity mix, weekly volume, and routing load by team.

The fourth chart is the one an operations manager would look at first. It shows how the workload lands across teams - Product Quality Engineering carrying the largest share, then Pharmacovigilance. Before this system existed, that distribution was invisible until someone counted it by hand.

The engine scorecard

ENGINE COMPARISON — SAME 40 COMPLAINTS, SAME ANSWER KEY			
METRIC	FULLY MANUAL	KEYWORD RULES	CLAUDE + HUMAN REVIEW
Minutes per complaint	12.0	5.0	4.1
Category accuracy	—	72%	98%
Product ID accuracy	—	98%	100%
Severity accuracy	—	42%	95%
Routing accuracy	—	65%	85%
Serious severity under-calls	—	15 of 40	1 of 40
Records still needing a human	100%	100%	70%
Both engines were run over the identical 40 complaints and scored against the same human-reviewed answer key. Every figure above is calculated and formatted by <code>scripts/assessment.py</code> dashboard using the same code that prints the <code>evaluate</code> table, so the two cannot disagree. The highlighted row is the one that matters most: an under-call is a harmful complaint filed as routine.			
HUMAN REVIEW QUEUE — HIGHEST PRIORITY FIRST			

Figure 8 - The two engines side by side, as shown on the live dashboard.

This is the same table that appears in sections 1 and 8 of this document, but here it is generated by the software rather than typed by an author. The green figures mark the better of the two engines on each row, and the highlighted row is the safety measure.

A small detail that matters more than it looks

The percentages on this screen are calculated AND formatted by the Python code, then passed to the browser as finished text.

The first version let the browser do its own rounding, and it immediately disagreed with the command-line output - 73% against 72%, because the two languages round a value of exactly 72.5 in opposite directions. A dashboard that contradicts the report it is supposed to evidence is worse than no dashboard, so the second formatter was removed entirely.

The human review queue

ID	SEV	PRI	PRODUCT	RE SUMMARY	MISSING INFORMATION	SUGGESTED NEXT ACTION	TEAM	CONF
C-1037	Critical	99	PenMate Insulin Pen	Pen delivered a suspected full cartridge in one press; patient found unresponsive with glucose 1.9 and hospitalised.	event_date pen_dose_setting insulin_type discharge_summary	Open MDR assessment within 24 hours and secure the pen for dose delivery testing.	Pharmacovigilance	0.97
C-1002	Critical	98	GlucoTrack G5 Blood Glucose Meter	Discordant glucose readings led to insulin dose and overnight emergency admission for severe hypoglycaemia.	event_date meter_lot_and_serial test_strip_lot discharge_summary	Open MDR assessment within 24 hours and call patient today.	Pharmacovigilance	0.95
C-1025	Critical	97	InfuLine 300 Infusion Pump	Suspected free-flow bolus on set removal; neonate became fluid overloaded and required transfer to higher level care.	event_date pump_serial_number set_lot infusate_and_volume patient_outcome	Open MDR assessment immediately and collect pump and set for joint investigation.	Product Quality Engineering	0.96
C-1007	Critical	96	Cardiozan 40mg Tablets	Facial swelling and diffuse rash after two weeks on Cardiozan; treated with steroids at urgent care and drug stopped.	event_date patient_age_and_sex concomitant_medications outcome_at_follow_up	Open expedited pharmacovigilance case and request follow-up outcome.	Pharmacovigilance	0.94
C-1030	Critical	95	SureSeal Surgical Stapler	Post-operative infection linked to compromised sterile pouch; patient returned to theatre for washout and IV antibiotics.	event_date operative_and_microbiology_reports pouch_photographs patient_outcome	Open MDR assessment and joint sterility investigation with the retained packaging.	Pharmacovigilance	0.86
C-1040	Critical	94	InfuLine 300 Infusion Pump	Air-in-line alarm failed to trigger with visible air column in the set; caught visually before reaching the patient.	event_date pump_serial_number set_lot alarm_test_results	Quarantine unit and test air-in-line detection across the installed fleet.	Product Quality Engineering	0.93
C-1013	High	88	SureSeal Surgical Stapler	Stapler misfired during bowel resection leaving incomplete staple line; surgeon oversewed and procedure extended 25 minutes.	event_date device_return_status operative_note patient_follow_up	Retrieve device immediately and open MDR assessment for intraoperative failure.	Product Quality Engineering	0.93
C-1029	High	88	Cardiozan 40mg Tablets	Two patients on one lot showed marked creatinine rise within two weeks; one admitted for rehydration.	event_date baseline_renal_function concomitant_medications patient_outcomes	Open pharmacovigilance cases and initiate lot-level signal review.	Pharmacovigilance	0.92
C-	High	87	AeroRest CPAP Unit	Unit shut off twice overnight without alarm; patient woke gasping	event_date	Replace unit immediately and retrieve original	Product Quality	0.9

Figure 9 - The working queue, ordered with the most serious complaint first.

This is the screen a complaint handler would actually use. Each row shows the AI's draft: a severity badge, a priority score, the identified product, a one-sentence summary, the list of missing facts, a suggested next action, the proposed team, and the AI's confidence.

The top row is complaint C-1037 - an insulin pen suspected of delivering a full cartridge in one press, with the patient found unresponsive and hospitalised. The AI graded it Critical, scored it 99 for priority, listed four facts that must be chased, and recommended opening a regulatory assessment within 24 hours. A handler can act on that in seconds rather than spending twelve minutes reconstructing it.

10. Limitations, risks and requirements

This section is deliberately blunt. A capstone that lists no weaknesses has not been examined properly.

Limitations of the evidence

Limitation	Why it matters and what it would take to fix
The dataset is synthetic, and one author wrote the complaints, the answer key AND the AI output in a single session	This means the accuracy figures measure internal consistency, not independent agreement. They are illustrative, not validated. Fixing it requires real complaint text and at least two independent reviewers producing the answer key.
Only 40 records	Percentages on 40 records have wide error bars. A single reclassification moves any figure by 2.5 points.
The 12-minute manual baseline is assumed, not measured	It is the single most load-bearing number in the business case. Time 30 real complaints before funding anything.
The keyword comparison engine was written by the same person as the AI prompt	A more sympathetic engineer might build a better rules engine. The gap would narrow, though the severity gap would not close - keyword matching cannot read narrative.

Risks in operation

Risk	Mitigation built into the design
The AI gives different answers on different runs	Temperature is set to zero, the prompt is version-controlled, and the AI draft is stored separately from the human decision so any drift is visible and auditable.
Handlers stop reading and just click Accept	This is the most likely real-world failure. Mitigation: track the accept-without-edit rate. If it exceeds 95% while a sample audit disagrees, the review has become a rubber stamp and must be re-designed.
Patient data leaves the company through the AI service	Requires a zero-retention API configuration, a signed data processing agreement, regional endpoint selection for EU data, and a de-identification pass at intake.
Malicious or confusing text in a complaint manipulates the AI	The complaint text is untrusted input. The output is constrained to a fixed schema with closed lists, so the worst case is a wrong classification, not an arbitrary instruction being followed.
The model or prompt is changed and quality silently degrades	The agreement rate is monitored continuously. Below 85%, auto-routing is disabled automatically.

Requirements before this could be used for real

- A validated, version-controlled prompt with a formal change-control procedure, treated as regulated documentation.
- A zero-retention agreement with the AI provider and a signed data processing agreement.
- De-identification of direct identifiers at the point of intake.
- A defined privacy-incident route, because complaint text sometimes discloses a third party's data - this project's own C-1035 record does exactly that.
- A documented human review procedure, with named roles and a defined escalation path.
- Retention of both the AI draft and the human decision for the full record retention period.
- A 90-day shadow-mode pilot before any automatic routing is enabled.

11. What would have to be true before this goes live

The recommended path is a 90-day pilot in shadow mode. The AI assesses every incoming complaint, handlers continue to work entirely by hand, and neither sees the other's answer. At the end of 90 days the two are compared.

Go-live conditions

Condition	Threshold
Serious under-call rate	Zero on the pilot set, with no exceptions
Agreement rate with human reviewers	At or above 85%, sustained across the full 90 days
Measured manual baseline	Confirms handling time above 9 minutes, or the economics fail
Handler acceptance	Reviewers report the draft saves time rather than creating checking work
Privacy and data controls	All requirements in section 10 signed off before any real text is sent

The honest recommendation

Fund the pilot, not the build.

The efficiency case is thin and rests on an unmeasured assumption. The safety case is strong but rests on 40 synthetic records. Ninety days of shadow running costs very little and would replace both weaknesses with evidence.

If the under-call finding survives contact with real complaints, the project justifies itself on safety alone and the time saving becomes a bonus rather than the argument.

12. Appendix

Repository contents

Path	What it holds
data/complaints_raw.csv	The 40 synthetic complaints
data/products.csv	The 10-product catalogue
data/labels_gold.csv	The answer key used to score both engines
data/assessment_ai.csv	The AI arm output - the main artifact. Ships pre-generated so the comparison can be reproduced offline.
data/assessment_baseline.csv	The keyword rules output - the comparison arm
prompts/assessment_prompt.md	The prompt, with the design rationale for every choice
scripts/assessment.py	The pipeline: run, evaluate, dashboard
sql/01_schema.sql	Database schema - 3 tables and 2 reporting views
dashboard/index.html	The static dashboard - charts, the engine scorecard, and the review queue
docs/business-case.md	Assumptions, arithmetic, and what would falsify the case
docs/human-in-the-loop.md	The review procedure and escalation rules
docs/risks-and-limitations.md	Risks, mitigations and deployment requirements

How to reproduce the numbers in this document

Part A - the offline path (no dependencies, no API key, no internet)

These three commands rebuild the comparison arm, re-score both engines against the answer key, and refresh the dashboard. Every accuracy figure, every under-call count and the handling-time model in section 8 are printed by the second command.

```
cd scripts
python assessment.py run --engine keyword    # rebuild the keyword comparison arm
python assessment.py evaluate               # print the full metric table
python assessment.py dashboard              # refresh the dashboard data
```

Then open dashboard/index.html in any browser. Python 3 is the only requirement; nothing needs to be installed. The dashboard charts load a charting library from the internet, so they need a connection - the summary panel and the review queue work offline.

Part B - regenerating the AI arm

The three commands above do NOT regenerate the AI output. The file `data/assessment_ai.csv` ships with the project as a saved artifact, and the scoring step reads it from disk. This is deliberate: it means anyone can reproduce and audit the comparison without an API key, an account, or an internet connection.

To rebuild the AI arm from scratch, two things must be in place first - the Anthropic client library, and an API key held in an environment variable rather than written into any file:

```
pip install anthropic      # the only third-party dependency in the project
$env:ANTHROPIC_API_KEY = "sk-ant-..."    # never commit a key to the repository
python assessment.py run --engine claude    # one API call per complaint, 40 in total
python assessment.py evaluate               # re-score with the newly generated arm
python assessment.py dashboard             # refresh the dashboard again
```

Three honest warnings about running Part B

1. It overwrites `data/assessment_ai.csv`. Take a copy first if you want to keep the arm the figures in this document were calculated from.
2. The results may differ. Temperature is set to zero, which makes the model as repeatable as it can be, but it is not a guarantee. If the output changes, the accuracy table in sections 1 and 8 must be updated to match - leaving the document and the data disagreeing would be worse than not re-running at all.
3. It costs money and needs internet. Roughly \$0.50 of API credit for all 40 complaints, at about one cent each.

This split is also why section 10 lists the single-author provenance of the AI arm as the first limitation. Part A is fully reproducible by anyone. Part B is what would replace the current AI output with an independently generated one.

End of document